

In many cases, technical papers include the analysis on the key question: “How secure is this cryptographic algorithm?” or rather “Why this cryptographic algorithm is secure?”. In this section, we will shortly describe the notation and typical construction for justifying the security of cryptographic algorithms.

Typically, the cryptographic security is defined in a form of a game between the adversary (who we call $\mathcal{A}$) and the challenger (who we call $\mathcal{Ch}$). The adversary is trying to break the security of the cryptographic algorithm using arbitrary (but still efficient) protocol, while the challenger is following a simple, fixed protocol. The game is played in a form of a challenge, where the adversary is given some information and is asked to perform some task. The security of the cryptographic algorithm is defined based on the probability of the adversary to win the game.

0.1 Cipher Semantic Security

Let us get into specifics. Suppose that we want to specify that the encryption scheme is secure. Recall that cipher $\mathcal{E} = (E, D)$ over the space $(\mathcal{K}, \mathcal{M}, \mathcal{C})$ (here, $\mathcal{K}$ is the space containing all possible keys, $\mathcal{M}$ – all possible messages and $\mathcal{C}$ – all possible ciphers) consists of two efficiently computable methods:

- $E : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{C}$ – encryption method, that based on the provided message $m \in \mathcal{M}$ and key $k \in \mathcal{K}$ outputs the cipher $c = E(k, m) \in \mathcal{C}$.
- $D : \mathcal{K} \times \mathcal{C} \rightarrow \mathcal{M}$ – decryption method, that based on the provided cipher $c \in \mathcal{C}$ and key $k \in \mathcal{K}$ outputs the message $m = D(k, c) \in \mathcal{M}$.

Of course, we require the **correctness**:

$$(\forall k \in \mathcal{K}) (\forall m \in \mathcal{M}) : \{D(k, E(k, m)) = m\}$$

Now let us play the following game between adversary $\mathcal{A}$ and challenger $\mathcal{Ch}$:

1. $\mathcal{A}$ picks any two messages $m_0, m_1 \in \mathcal{M}$ on his choice.
2. $\mathcal{Ch}$ picks a random key $k \leftarrow \mathcal{K}$ and random bit $b \leftarrow \{0, 1\}$ and sends the cipher $c = E(k, m_b)$ to $\mathcal{A}$.
3. $\mathcal{A}$ is trying to guess the bit b by using the cipher c .
4. $\mathcal{A}$ outputs the guess $\hat{b}$.

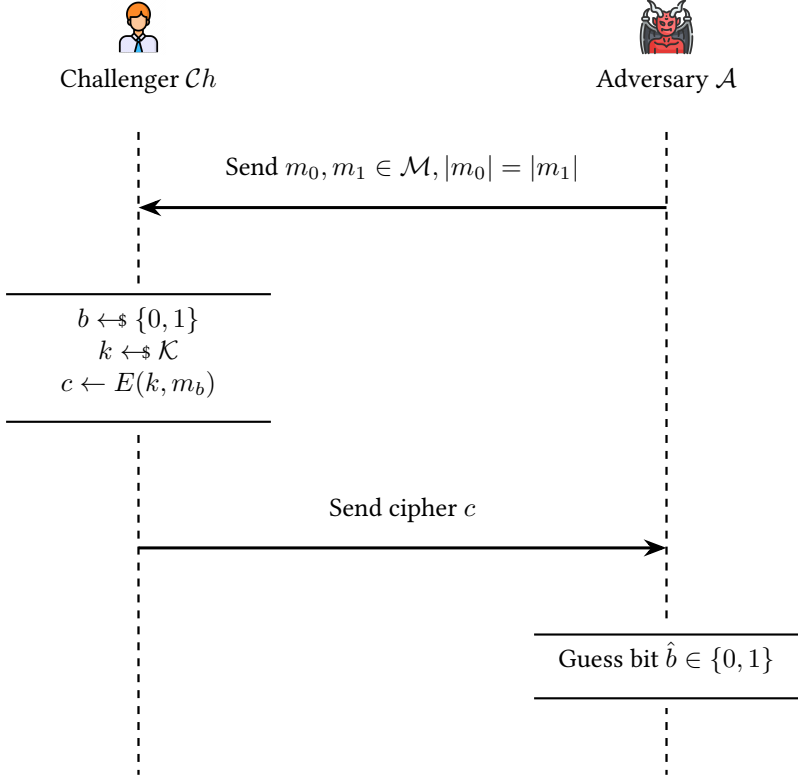


Figure 0.1: The game between the adversary $\mathcal{A}$ and the challenger $\mathcal{Ch}$ for defining the semantic security.

Now, what should happen if our encryption scheme is secure? The adversary should not be able to guess the bit b with a probability significantly higher than $1/2$ (a random guess). Formally, define the **advantage** of the adversary $\mathcal{A}$ as:

$$\text{SSAdv}[\mathcal{E}, \mathcal{A}] := \left| \Pr[\hat{b} = b] - \frac{1}{2} \right|$$

We say that the encryption scheme is **semantically secure**¹ if for any efficient adversary $\mathcal{A}$ the advantage $\text{SSAdv}[\mathcal{A}]$ is negligible. In other words, the adversary cannot guess the bit b with a probability significantly higher than $1/2$.

Now, what negligible means? Let us give the formal definition!

Definition 0.1. A function $f : \mathbb{N} \rightarrow \mathbb{R}$ is called **negligible** if for all $c \in \mathbb{R}_{>0}$ there exists $n_c \in \mathbb{N}$ such that for any $n \geq n_c$ we have $|f(n)| < 1/n^c$.

¹This version of definition is called a **bit-guessing** version.

The alternative definition, which is probably easier to interpret, is the following.

Theorem 0.2. A function $f : \mathbb{N} \rightarrow \mathbb{R}$ is **negligible** if and only if for any $c \in \mathbb{R}_{>0}$, we have

$$\lim_{n \rightarrow \infty} f(n)n^c = 0$$

Example 0.1. The function $f(n) = 2^{-n}$ is negligible since for any $c \in \mathbb{R}_{>0}$ we have

$$\lim_{n \rightarrow \infty} 2^{-n}n^c = 0$$

The function $g(n) = \frac{1}{n!}$ is also negligible for similar reasons.

Example 0.2. The function $h(n) = \frac{1}{n}$ is not negligible since for $c = 1$ we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \times n = 1 \neq 0$$

Well, that is weird. For some reason we are considering a function the depends on some natural number n , but what is this number?

Typically, when defining the security of the cryptographic algorithm, we are considering the security parameter λ (e.g., the length of the key). The function is negligible if the probability of the adversary to break the security of the cryptographic algorithm is decreasing with the increasing of the security parameter λ . Moreover, we require that the probability of the adversary to break the security of the cryptographic algorithm is decreasing faster than any polynomial function of the security parameter λ .

So all in all, we can define the semantic security as follows.

Definition 0.3. The encryption scheme $\mathcal{E}$ with a security paramter $\lambda \in \mathbb{N}$ is **semantically secure** if for any efficient adversary $\mathcal{A}$ we have:

$$\left| \Pr \left[b = \hat{b} \mid \begin{array}{l} m_0, m_1 \leftarrow \mathcal{A}, k \leftarrow \mathcal{K}, b \leftarrow \{0, 1\} \\ c \leftarrow E(k, m_b) \\ \hat{b} \leftarrow \mathcal{A}(c) \end{array} \right] - \frac{1}{2} \right| < \text{negl}(\lambda)$$

Do not be afraid of such complex notation, it is quite simple. Notation $\Pr[A \mid B]$ means “the probability of A , given that B occurred”. So our inner probability is read as “the probability that the guessed bit $\hat{b}$ equals b given the setup on the right”. Then, on the right we define the setup: first we generate two messages $m_0, m_1 \in \mathcal{M}$, then we choose a random bit b and a key k , cipher the message m_b , send it to the adversary and the adversary, based on provided cipher,

gives $\hat{b}$ as an output. We then claim that the probability of the adversary to guess the bit b is close to $1/2$.

Let us see some more examples of how to define the security of certain cryptographic objects.

0.2 Message recovery attacks

Essentially, message recovery attacks [?, chapter 2] are types of attacks that, from given ciphertext, recover the message with significantly better probability than random guessing, which is obviously $1/|\mathcal{M}|$. Of course, any reasonable notion of security should rule out such an attack, and indeed, semantic security does.

Although this might seem intuitively obvious, we provide a formal proof to clarify the reasoning. One of the reasons for doing this is to thoroughly demonstrate the concept of a *security reduction*, which is the primary method used to assess the security of systems.

Let us play the following attack game message recovery between adversary $\mathcal{A}$ and challenger $\mathcal{Ch}$. For a given cipher $\mathcal{E} = (E, D)$, defined over $\mathcal{K}, \mathcal{M}, \mathcal{C}$, and for a given adversary $\mathcal{A}$, the attack game proceeds as follows:

- The challenger computes $m \leftarrow \$ \mathcal{M}$, $k \leftarrow \$ \mathcal{K}$, $c \leftarrow \$ E(k, m)$ and sends c to the adversary.
- the adversary output a message $\hat{m} \in \mathcal{M}$.

We say that $\mathcal{A}$ wins the game in this case, and we define $\mathcal{A}$'s message recovery advantage with respect to $\mathcal{E}$ as:

$$\text{MRadv}[\mathcal{E}, \mathcal{A}] := \left| \Pr[\hat{m} = m] - \frac{1}{|\mathcal{M}|} \right|.$$

Definition 0.4 (Security against message recovery). A cipher $\mathcal{E}$ is secure against message recovery if for all efficient adversaries $\mathcal{A}$, the value $\text{MRadv}[\mathcal{E}, \mathcal{A}]$ is negligible.

Theorem 0.5. Let $\mathcal{E} = (E, D)$ be a cipher defined over $(\mathcal{K}, \mathcal{M}, \mathcal{C})$. If $\mathcal{E}$ is semantically secure then $\mathcal{E}$ is secure against message recovery.

► Let us prove by assuming the opposite statement. In other words, if $\mathcal{E}$ is not secure against message recovery, then $\mathcal{E}$ is not semantically secure.

Assume that $\mathcal{E}$ is not secure against message recovery. We have an efficient adversary $\mathcal{A}$ who has a significant advantage in the message recovery game and an efficient adversary $\mathcal{B}$ who is trying to compromise $\mathcal{E}$. The proof idea is to use an efficient adversary $\mathcal{A}$ as a "black box" (or oracle machine). We construct $\mathcal{B}$ as follows:

1. Adversary $\mathcal{B}$ generates $m_0, m_1 \in \mathcal{M}$ and sends them to the semantic security challenger.
2. The semantic security challenger returns ciphertext c to adversary $\mathcal{B}$. Since $\mathcal{B}$ can use the message recovery oracle, it sends c to $\mathcal{A}$.
3. $\mathcal{A}$ returns $\hat{m}$. If $\hat{m} = m_0$, then $b = 0$ otherwise, $b = 1$.
4. Sends b to the semantic security challenger.

Intuitively, this implies that $\text{SSadv}[\mathcal{E}, \mathcal{B}]$ is exactly equal to $\text{MRadv}[\mathcal{E}, \mathcal{A}]$. ◀

This means that if an adversary is capable of successfully winning the message recovery game, then they can also effectively win the semantic security game. In essence, security against message recovery is a stronger property than semantic security.

Our motivation for demonstrating proof is to illustrate in detail the notion of a security reduction, which is the main technique used to reason about the security of systems.

Security reduction is typically applied to certain computational problems. To be more precise, we will now explain how security reduction operates in general.

Definition 0.6 (Security Reduction).

1. Let $\mathcal{A}$ be an efficient adversary who is trying to compromise the cryptosystem $\mathcal{E}$.
2. We construct an efficient algorithm $\mathcal{A}'$, which we call the **reduction**, that will solve the computationally hard problem $\mathcal{X}$ while using an oracle access to $\mathcal{A}$. For any input x of the problem $\mathcal{X}$, the algorithm $\mathcal{A}'$ generates the input of the cryptosystem $\mathcal{E}$ for $\mathcal{A}$. Once it wins the corresponding experiment, $\mathcal{A}'$ solves $\mathcal{X}$ for input x with a certain non-negligible probability $\mu(n)$. Typically, such probability turns out to be bounded below by $1/n^c$, $c \in \mathbb{N}$.
3. Thus, the algorithm $\mathcal{A}'$ solves $\mathcal{X}$ with the non-negligible probability $\varepsilon(n)\mu(n)$ (or typically $\varepsilon(n)/n^c$). This quantity is non-negligible if $\varepsilon(n)$ is assumed to be non-negligible.
4. Since the problem $\mathcal{X}$ is computationally hard, we get a contradiction. It follows that no attacker $\mathcal{A}$ can effectively crack $\mathcal{E}$, i.e., this cryptosystem is computationally strong.

0.3 Discrete Logarithm Assumption

Now, let us define the fundamental assumption used in cryptography formally: the **Discrete Logarithm Assumption** (DL) [?, chapter 16].

Definition 0.7. Assume that $\mathbb{G}$ is a cyclic group of prime order r generated by $g \in \mathbb{G}$. Define the following game:

1. Both challenger $\mathcal{Ch}$ and adversary $\mathcal{A}$ take a description $\mathbb{G}$ as an input:

- order r and generator $g \in \mathbb{G}$.
- 2. $\mathcal{Ch}$ computes $\alpha \leftarrow \mathbb{Z}_r, u \leftarrow g^\alpha$ and sends $u \in \mathbb{G}$ to $\mathcal{A}$.
- 3. The adversary $\mathcal{A}$ outputs $\hat{\alpha} \in \mathbb{Z}_r$.

We define $\mathcal{A}$'s **advantage in solving the discrete logarithm problem in $\mathbb{G}$** , denoted as $\text{DLadv}[\mathcal{A}, \mathbb{G}]$, as the probability that $\hat{\alpha} = \alpha$.

Definition 0.8. The **Discrete Logarithm Assumption** holds in the group $\mathbb{G}$ if for any efficient adversary $\mathcal{A}$ the advantage $\text{DLadv}[\mathcal{A}, \mathbb{G}]$ is negligible.

Informally, this assumption means that given u , it is very hard to find α such that $u = g^\alpha$. But now we can write down this formally!

0.4 Computational Diffie-Hellman

Another fundamental problem in cryptography is the **Computational Diffie-Hellman** (CDH) problem [?, chapter 16]. It states that given g^α, g^β it is hard to find $g^{\alpha\beta}$. This property is frequently used in the construction of cryptographic protocols such as the Diffie-Hellman key exchange.

Let us define this problem formally.

Definition 0.9. Let $\mathbb{G}$ be a cyclic group of prime order r generated by $g \in \mathbb{G}$. Define the following game:

- 1. Both challenger $\mathcal{Ch}$ and adversary $\mathcal{A}$ take a description $\mathbb{G}$ as an input: order r and generator $g \in \mathbb{G}$.
- 2. $\mathcal{Ch}$ computes $\alpha, \beta \leftarrow \mathbb{Z}_r, u \leftarrow g^\alpha, v \leftarrow g^\beta, w \leftarrow g^{\alpha\beta}$ and sends $u, v \in \mathbb{G}$ to $\mathcal{A}$.
- 3. The adversary $\mathcal{A}$ outputs $\hat{w} \in \mathbb{G}$.

We define $\mathcal{A}$'s **advantage in solving the computational Diffie-Hellman problem in $\mathbb{G}$** , denoted as $\text{CDHadv}[\mathcal{A}, \mathbb{G}]$, as the probability that $\hat{w} = w$.

Definition 0.10. The **Computational Diffie-Hellman Assumption** holds in the group $\mathbb{G}$ if for any efficient adversary $\mathcal{A}$ the advantage $\text{CDHadv}[\mathcal{A}, \mathbb{G}]$ is negligible.

0.5 Decisional Diffie-Hellman

Now, we loosen the requirements a bit. The **Decisional Diffie-Hellman** (DDH) problem [?, chapter 16] states that given $g^\alpha, g^\beta, g^{\alpha\beta}$ it is “hard” to distinguish $g^{\alpha\beta}$ from a random element in $\mathbb{G}$. Formally, we define this problem as follows.

Definition 0.11. Let $\mathbb{G}$ be a cyclic group of prime order r generated by $g \in \mathbb{G}$. Define the following game:

1. Both challenger $\mathcal{Ch}$ and adversary $\mathcal{A}$ take a description $\mathbb{G}$ as an input: order r and generator $g \in \mathbb{G}$.
2. $\mathcal{Ch}$ computes $\alpha, \beta, \gamma \leftarrow \mathbb{Z}_r, u \leftarrow g^\alpha, v \leftarrow g^\beta, w_0 \leftarrow g^{\alpha\beta}, w_1 \leftarrow g^\gamma$. Then, $\mathcal{Ch}$ flips a coin $b \leftarrow \{0, 1\}$ and sends u, v, w_b to $\mathcal{A}$.
3. The adversary $\mathcal{A}$ outputs the predicted bit $\hat{b} \in \{0, 1\}$.

We define $\mathcal{A}$'s **advantage in solving the Decisional Diffie-Hellman problem in $\mathbb{G}$** , denoted as $\text{DDHadv}[\mathcal{A}, \mathbb{G}]$, as

$$\text{DDHadv}[\mathcal{A}, \mathbb{G}] := \left| \Pr[b = \hat{b}] - \frac{1}{2} \right|$$

Now, let us break this assumption for some quite generic group! Consider the following example.

Theorem 0.12. Suppose that $\mathbb{G}$ is a cyclic group of an even order. Then, the Decision Diffie-Hellman Assumption does not hold in $\mathbb{G}$. In fact, there is an efficient adversary $\mathcal{A}$ that can distinguish $g^{\alpha\beta}$ from a random element in $\mathbb{G}$ with an advantage $1/4$.

Proof. If $|\mathbb{G}| = 2n$ for $n \in \mathbb{N}$, it means that we can split the group into two subgroups of order n , say, $\mathbb{G}_1$ and $\mathbb{G}_2$. The first subgroup consists of elements in a form g^{2k} , while the second subgroup consists of elements in a form g^{2k+1} .

Now, if we could efficiently determine, based on group element $g \in \mathbb{G}$, whether $g \in \mathbb{G}_1$ or $g \in \mathbb{G}_2$, we essentially could solve the problem. Fortunately, there is such a method! Consider the following lemma.

Lemma 0.13. Suppose $u = g^\alpha$. Then, α is even if and only if $u^n = 1$.

Proof. If α is even, then $\alpha = 2\alpha'$ and thus

$$u^n = (g^{2\alpha'})^n = g^{2n\alpha'} = (g^{2n})^{\alpha'} = 1^{\alpha'} = 1$$

Conversely, if $u^n = 1$ then $u^{\alpha n} = 1$, meaning that $2n \mid \alpha n$, implying that α is even. Lemma is proven.

Now, we can construct our adversary $\mathcal{A}$ as follows. Suppose $\mathcal{A}$ is given (u, v, w) . Then,

1. Based on u , get the parity of α , say $p_\alpha \in \{\text{even}, \text{odd}\}$.
2. Based on v , get the parity of β , say $p_\beta \in \{\text{even}, \text{odd}\}$.
3. Based on w , get the parity of γ , say $p_\gamma \in \{\text{even}, \text{odd}\}$.
4. Calculate $p'_\gamma \in \{\text{even}, \text{odd}\}$ — parity of $\alpha\beta$.
5. Return $\hat{b} = 0$ if $p'_\gamma = p_\gamma$, and $\hat{b} = 1$, otherwise.

Suppose γ is indeed $\alpha \times \beta$. Then, condition $p'_\gamma = p_\gamma$ will always hold. If γ is a random element, then the probability that $p'_\gamma = p_\gamma$ is $1/2$. Therefore, the probability that $\mathcal{A}$ will guess the bit b correctly is $3/4$, and the advantage is $1/4$ therefore. ■

Why is this necessary? Typically, it is impossible to prove the predicate “for every efficient adversary $\mathcal{A}$ this probability is negligible” and therefore we need to make assumptions, such as the Discrete Logarithm Assumption or the Computational Diffie-Hellman Assumption. In turn, proving the statement “if X is secure then Y is also secure” is manageable and does not require solving any fundamental problems. So, for example, knowing that the probability of the adversary to break the Diffie-Hellman assumption is negligible, we can prove that the Diffie-Hellman key exchange is secure.

0.6 Exercises

Exercise 1. Which of the following functions is **negligible**?

- a) $\mu(\lambda) = \lambda^{-2^{10}}$.
- b) $\mu(\lambda) = 5^{-\log \lambda}$.
- c) $\mu(\lambda)$ is λ^{-1000} for all odd λ and $2^{-\lambda}$ for all even λ .
- d) $\mu(\lambda) = \lambda^{100} e^{-\sqrt{\lambda}}$.

Exercise 2. Let $\mathcal{M} = \mathcal{C} = \mathcal{K} = \mathbb{Z}_p^\times$ for a prime p . Define the encryption scheme $\mathcal{E} = (E, D)$ as follows:

$$E(k, m) = k \cdot m \pmod{p}, \quad D(k, c) = k^{-1} \cdot c \pmod{p}$$

Is the scheme $\mathcal{E}$ perfectly secure? (*Perfect security means that adversary's advantage is exactly 0*).

- a) Yes.
- b) No, because only the one-time pad can be perfectly secret.
- c) No, because $|\mathcal{K}|$ must be greater than $|\mathcal{M}|$.
- d) No, because there is a simple strategy for an adversary to find the message given the ciphertext.

Exercise 3. For an encryption scheme $\mathcal{E}$, an adversary can guess the last bit of a message m from its ciphertext c with probability $\frac{1}{2} + \varepsilon$ via a function $f(c)$. The adversary plays the IND-CPA game by choosing m_0 with last bit 0 and m_1 with last bit 1. Upon receiving ciphertext c_b , the adversary returns $\hat{b} = f(c_b)$. What is the adversary's advantage, defined as $\text{SSadv}[\mathcal{A}, \mathcal{E}] := 2 \left| \Pr[\hat{b} = b] - \frac{1}{2} \right|$?

- a) 2ε .
- b) $\frac{1}{2} + \varepsilon$.
- c) ε^2 .
- d) $1 - 2\varepsilon$.

Exercise 4. Let $\mathcal{E} = (E, D)$ be a semantically secure encryption scheme where $\mathcal{M} = \mathcal{C} = \{0, 1\}^\lambda$. Which of the following schemes are semantically secure? (*Multiple options may be correct.*)

Note: $\text{parity}(m)$ is 1 if the number of ones in m is odd, and 0 otherwise. $\parallel$ is concatenation, and $\text{reverse}(m)$ reverses the bit string m .

- a) $E'(k, m) = 0 \parallel E(k, m)$.
- b) $E'(k, m) = E(k, m) \parallel \text{parity}(m)$.
- c) $E'(k, m) = \text{reverse}(E(k, m))$.
- d) $E'(k, m) = E(k, \text{reverse}(m))$.
- e) $E'(k, m) = E(k, m) \parallel k$.
- f) $E'((k, k'), m) = E(k, m) \parallel E(k', m)$.

Exercise 5. Let $G(s)$ be a secure pseudorandom generator (PRG) that generates λ bits. Which of the following generators are secure? (*Multiple options may be correct.*)

- a) $G'(s) = G(s) \parallel 0$.
- b) $G'(s_1 \parallel s_2) = G(s_1) \oplus G(s_2)$.
- c) $G'(s) = G(s) \oplus 1^\lambda$, where 1^λ is a string of λ ones.
- d) $G'(s)$ is the first $\lambda/2$ bits of $G(s)$.
- e) $G'(s) = G(s) \parallel G(s)$.
- f) $G'(s_1 \parallel s_2) = (s_1, G(s_2))$.
- g) $G'(s) = G(0)$.

Exercises 6-8. PRG Security and Statistical Tests

Let $G : \mathcal{K} \rightarrow \{0, 1\}^\lambda$ be a pseudorandom generator. We want to test if G is secure using a function $\mathcal{A} : \{0, 1\}^\lambda \rightarrow \{0, 1\}$, which we call a *statistical test*. The advantage of $\mathcal{A}$ is defined as:

$$\text{PRGadv}[\mathcal{A}, G] = \left| \Pr_{k \leftarrow \mathcal{K}}[\mathcal{A}(G(k)) = 1] - \Pr_{r \leftarrow \mathcal{S}\{0, 1\}^\lambda}[\mathcal{A}(r) = 1] \right|$$

An alternative definition for the security of G is that $\text{PRGadv}[\mathcal{A}, G]$ is negligible for all efficient algorithms $\mathcal{A}$.

Exercise 6. Consider two trivial statistical tests: (a) $\mathcal{A}_0(x) = 0$ for all x and (b) $\mathcal{A}_1(x) = 1$ for all x . Which of the following statements is true?

- a) $\text{PRGadv}[\mathcal{A}_0, G] = \text{PRGadv}[\mathcal{A}_1, G] = 0$.
- b) $\text{PRGadv}[\mathcal{A}_0, G] = \text{PRGadv}[\mathcal{A}_1, G] = 1$.
- c) $\text{PRGadv}[\mathcal{A}_0, G] = \text{PRGadv}[\mathcal{A}_1, G] = \frac{1}{2}$.
- d) $\text{PRGadv}[\mathcal{A}_0, G] = 0, \text{PRGadv}[\mathcal{A}_1, G] = 1$.

Exercise 7. Let a PRG $G : \mathcal{K} \rightarrow \{0, 1\}^\lambda$ satisfy the condition that the last

bit of $G(k)$ is one for $2/3$ of the keys in $\mathcal{K}$. Consider the statistical test $\mathcal{A}(x)$ which outputs 1 if the last bit of x is one, and 0 otherwise. What is the advantage $\text{PRGadv}[\mathcal{A}, G]$?

- a) 1. b) $1/2$. c) $1/3$. d) $2/3$. e) $1/6$.

Exercise 8. Let $G : \mathcal{K} \rightarrow \{0, 1\}^\lambda$ be a secure PRG. Define a new generator $G' : \mathcal{K} \times \mathcal{K} \rightarrow \{0, 1\}^\lambda$ as:

$$G'(k_1, k_2) = G(k_1) \wedge G(k_2), \quad \text{where } \wedge \text{ is bitwise AND.}$$

Consider the statistical test $\mathcal{A}(x) = \{\text{the first bit of } x\}$. What is the advantage $\text{PRGadv}[\mathcal{A}, G']$?

- a) G' is secure.
b) $1/2$.
c) $1/3$.
d) $1/4$.
e) 1.